Name: Key ____________ / 25

Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

There are 6 questions worth 25 points on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. **[6 points]** Given functions $f(x) = x^3 - 3x + 1$ and $g(x) = 2^x$, compute the following and **simplify** your answers:

$f(2) = 8 - 6 + 1 = 3$

a. $f(2) + g(2)$

$= 3 + 4 = 7$

7

b. $f(0) - g(0)$

$= 1 - 1 = 0$

0

c. $g(f(1))$

$= g(-1) = 2^{-1} = \frac{1}{2}$

$\frac{1}{2}$

2. **[3 points]** Write an equation of the line passing through the points $(1,2)$ and $(-1,5)$.

$m = \frac{5-2}{-1-1} = \frac{3}{-2} = -\frac{3}{2}$

$\boxed{y - 2 = -\frac{3}{2}(x-1)}$

or

$y = -\frac{3}{2}x + \frac{3}{2} + 2 \rightarrow \boxed{y = -\frac{3}{2}x + \frac{7}{2}}$

3. **[3 points]** Find all x-intercepts for the function $f(x) = 2x^2 - 7x + 5$.

$2x^2 - 7x + 5 = 0$

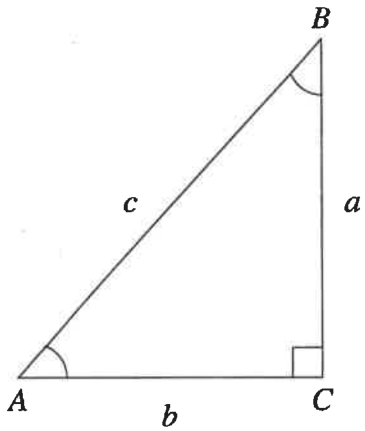
$(2x-5)(x-1) = 0 \rightarrow x = 1, \frac{5}{2}$

$\boxed{x\text{-intercepts are } (1,0) \text{ and } (\tfrac{5}{2}, 0)}$

4. [4 points] In the right triangle below, suppose $a = 5$ and $c = \sqrt{34}$.

$5^2 + b^2 = 34 \rightarrow b^2 = 9$

$b = 3$



$b = 3$

Calculate the following:

$\sin(A) = \frac{5}{\sqrt{34}} = \frac{5\sqrt{34}}{34}$

$\cos(A) = \frac{3}{\sqrt{34}} = \frac{3\sqrt{34}}{34}$

$\tan(A) = \frac{5}{3}$

5. [3 points] Evaluate the following:

a. $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ **b.** $\tan\left(\frac{5\pi}{6}\right) = \frac{-\sqrt{3}}{3}$ **c.** $\sec(\pi) = -1$

6. [6 points] State the domain and range of the following functions. Use interval notation.

a. $h(x) = \ln(x)$

domain: $(0, \infty)$ range: $(-\infty, \infty)$

b. $g(x) = 3\cos(x) + 1$

domain: $(-\infty, \infty)$ range: $[-2, 4]$

c. $f(x) = -\sqrt{x+1} - 5$

domain: $[-1, \infty)$ range: $(-\infty, -5]$